

Minseong Cho

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🏢 Softlab, Ajou University
📍 Suwon, South Korea
📁 Portfolio
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Research Interests

Active matter dynamics · Elastic Leidenfrost effect · Mechanics of Moisture-Driven Hair Frizz

Personal Statement

I am an M.S. student in Mechanical Engineering and a graduate researcher at Ajou University. My research focuses on understanding how soft and deformable systems move, adapt, and self-organize under nonequilibrium conditions, driven by energy input, environmental changes, and interactions with their surroundings.

Since my undergraduate years, I have developed a hands-on research approach by designing experiments, building quantitative image-analysis methods, and interpreting complex behaviors through mechanics and physics. I enjoy starting from subtle experimental observations, developing the analytical tools needed to characterize them, and uncovering the physical principles that govern their behavior.

As a researcher, I am particularly drawn to unfamiliar problems that require patience, technical flexibility, and careful interpretation. Rather than being defined by a single material or research topic, I aim to develop a versatile way of thinking that integrates experiments, quantitative analysis, and physical reasoning to reveal the physics behind complex phenomena across a wide range of systems.

Education

- **M.S. in Mechanical Engineering** Sep 2025 – Present
Ajou University, Suwon, South Korea
 - **Advisor:** Prof. Jonghyun Ha
- **B.S. in Mechanical Engineering** Mar 2020 – Aug 2025
Ajou University, Suwon, South Korea

Research Experience

- **Graduate Research Student – Softlab, Ajou University** Sep 2025 – Present
 - **Mechanics of Moisture-Driven Hair Frizz**
 - Focus:** Moisture-driven changes in strand morphology.
 - Techniques:** Centerline-based geometric analysis and simulation-assisted interpretation.
 - Advisor:** Prof. Jonghyun Ha, Dr. Yeonsu Jung
 - **Boundary-Driven Active Matter in Soft Hydrogels**
 - Focus:** Leidenfrost-induced dynamics in soft hydrogel systems.
 - Techniques:** Image-based tracking, curvature analysis, and visualization scripts.
 - Advisor:** Prof. Jonghyun Ha, Dr. Yeonsu Jung
- **Undergraduate Research Student – Softlab, Ajou University** Mar 2023 – Aug 2025
 - **Boundary-Driven Active Matter in Soft Hydrogels**
 - Focus:** Leidenfrost-related soft-matter dynamics.
 - Techniques:** Image-based tracking, quantitative analysis, and figure preparation.
 - Advisor:** Prof. Jonghyun Ha, Dr. Yeonsu Jung

Conferences and Presentations

- **APS Division of Fluid Dynamics Meeting (Houston, TX, USA)** Nov 2025
 - Leidenfrost-induced active matter dynamics of spherical hydrogels
- **Asian Congress of Fluid Mechanics (Seoul, South Korea)** Sep 2025
 - Leidenfrost-induced active matter dynamics of spherical hydrogels
- **KSME Annual Conference (Jeju, South Korea)** Nov 2024
 - Explosive motion of spherical hydrogel via the Leidenfrost effect

Awards & Honors

- **Dean's List** – Ajou University Spring 2024, Spring 2025
Recognized for outstanding academic performance (top 5% of the department).
- **Capstone Design Competition** – Ajou University June 2025
2nd Place · KRW 1,000,000 prize
Project: Compliant-mechanism-based knee assistive device for degenerative osteoarthritis prevention.
- **Gyeonggi-do Assistive Device Idea Competition** Nov 2025
3rd Place · KRW 300,000 prize
Organized by: Gyeonggi Province Rehabilitation Engineering Service Support Center (South Korea)
Project: Compliant-mechanism-based knee assistive device for older adults.
- **APS DFD Travel/Enabling Grant** 2025
US\$750 travel cost reimbursement
Organized by: APS Division of Fluid Dynamics, American Physical Society
Purpose: Travel support for participation in the APS Division of Fluid Dynamics Meeting.
- **Gallery of Fluid Motion** Apr 2026
Haecheon Choi GFM Award · KRW 400,000 prize
Organized by: The Korean Society of Mechanical Engineers, Fluid Engineering Division
Title: Explosive motion of spherical hydrogel via the Leidenfrost effect
Authors: Minseong Cho, Yeonsu Jung, Jonghyun Ha

Publications

- **Boundary-driven active matter and stochastic rectification via Leidenfrost impulses**
Minseong Cho, Yeonsu Jung, and Jonghyun Ha
In preparation

References

Prof. Jonghyun Ha

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Dr. Yeonsu Jung

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